

Sumit Das

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EDUCATION

- **Khulna University of Engineering & Technology** Khulna, Bangladesh
B.Sc. – Electronics and Communication Engineering; CGPA: 3.65/4.00 Jan 2020 – Sept 2025

RESEARCH INTERESTS

As a research-oriented professional and ECE graduate from KUET, my research journey began with a thesis focused on **Wireless Sensor Networks** by synergizing it with **Machine Learning** models to enhance energy-efficiency and network lifetime, supported by a strong history of project work in **IoT** ecosystems, **Computer Vision**, and fundamental Electrical Engineering. Recognizing the imperative need for secure and autonomous systems, I am actively broadening my research scope. I am currently investigating the intricacies of **Computer Networking** and **Cybersecurity**, with a specific emphasis on network security and threat mitigation. Parallel to this, I am engaged in studying **Large Language Models (LLMs)** and their potential integration into **Autonomous Unmanned Vehicle** frameworks. I view myself as a lifelong learner and a results-oriented researcher, striving to contribute meaningful innovations to the global engineering community.

SKILLS

- **Software Skills:** Cisco Packet Tracer, Ni Multisim, Proteus, MATLAB Simulink, OriginPro, CST Studio, Microsoft Office, Adobe Illustrator
- **Programming Languages:** C++, MATLAB, Python, HTML5, CSS3, PHP, SQL
- **Platforms:** Windows, Arduino, Raspberry Pi
- **Others:** Latex, Git

EXPERIENCE

ELITE Research Lab LLC Queens, NY, USA
Research Intern - Remote Jan 2026 – Present

- **Large Language Models:** I am currently working remotely to contribute to the lab's ongoing projects on Large Language Models and their applications.

Khulna University of Engineering & Technology Khulna, Bangladesh
Undergraduate Researcher, Dept. of Electronics and Communication Engineering Feb 2024 – Sept 2025

- **Wireless Sensor Network:** During my undergraduate thesis, I enhanced the network lifetime and energy efficiency of SEP and DEEC protocol by introducing improved probability, threshold equations and implementing advanced deployment and sleep-awake schedules of the nodes. The enhanced protocol nearly got 34% stability period improvement compared to the parent protocol.
- **Machine Learning integration with WSN:** Used Self-organizing maps for the deployment of nodes, which got around 7% better area coverage, and used clustering algorithms to organize the clusters in SEP protocol. Also used the random forest algorithm in deciding a secondary cluster-head for multi-hop routing.

PUBLICATIONS

1. **S. Das** and Md. K. Alom, "EVS-SEP: A Hybrid Efficient Zonal Stable Election Routing Protocol with SOM-Based Node Deployment for Heterogeneous Wireless Sensor Networks." Scientific African (under review, 2026) (**Undergraduate Thesis**)
2. S. M. Islam, **S. Das** and Md. K. Alom, "SA-DEEC: An Efficient Sleep-Awake Assisted DEEC Routing Protocol for Heterogeneous Wireless Sensor Network." (in preparation, 2026)

PROJECTS

Robotics & Computer Vision:

- i. **Target Retrieval Rover:** Built an autonomous line-following rover with a custom robotic claw. Utilized a hybrid architecture (**Arduino & Raspberry Pi**) with **OpenCV** for real-time color detection.
- ii. **Gesture based AI Keyboard and Volume Control:** Developed a contactless human-computer interface using **Python**, **OpenCV**, and **MediaPipe**. Integrated a virtual keyboard that simulates keystrokes via thumb-index pinch gestures (using **pynput**) and a dynamic volume controller that maps finger proximity to Windows Audio API commands for real-time system interaction.

- iii. **YOLO Object Detection & Tracking:** Developed a suite of real-time vision applications using **YOLOv8** and **SORT** for multi-object tracking. Implemented bi-directional traffic and pedestrian counters, and trained custom models for safety gear detection and real-time Poker Hand Analysis.
- iv. **Ai Football Analysis System:** Built a **YOLO**-based analysis tool to track football players and calculate speed/distance in meters. Utilized **K-Means** for team assignment, **Optical Flow** to handle camera movement, and Perspective Transformation to accurately map the field dimensions.

Computer Networks:

- i. **Campus Network:** Simulated a secure campus network (ECE, CSE, Admin) using **Cisco Packet Tracer**. Integrated DHCP, VLANs, Web/DNS servers, and RIP dynamic routing with ACL security.

Internet of Things:

- i. **IoT-Based Patient Monitoring System:** Developed a system using **ESP32** and **Arduino** to track vital signs (BPM, Temp, CO₂) and fall detection. Features real-time **MATLAB** visualization and automated SMS alerts.

Electronics & Simulation:

- i. **Variable DC Power Supply:** Designed a 220V AC to 5V–35V regulated DC supply. Implemented short-circuit and over/under-voltage protection ($> 30V$, $< 10V$) using **LM358** op-amps and transistor logic. Integrated **LM317T** regulators, relays, and SCRs for stable output. Simulated in **Proteus** and implemented on Veroboard.
- ii. **FIR Filter Design:** Designed a **MATLAB** bandpass filter (1500–2750 Hz) using Hanning/Hamming windows to suppress noise.
- iii. **UWB Vivaldi Antenna Design:** Designed and simulated an Ultra-Wideband (UWB) Vivaldi Antenna using **CST Studio Suite** (2.5–4 GHz). Achieved resonance at 3.2 GHz with a return loss of -47.06 dB, VSWR ≈ 1 , and 10.25 dBi directivity at 4 GHz. Optimized for narrow beamwidth (HPBW $< 70^\circ$) to enable precise wireless detection.

Web & Database:

- i. **TeliGari-BaZaar (E-commerce):** Developed an automotive e-commerce site using **PHP** and **MySQL** with secure authentication and persistent carts.
- ii. **Blood Bank Management DBMS:** Created an **SQL** database to automate blood inventory, expiration tracking, and donor availability dashboards.

Explore my **Github**: [SumitDas44](#) 

TRAINING

EDGE Project of ICT Division

KUET, Khulna, Bangladesh

Trainee, Machine Learning with Python

Dec 2024 – Feb 2025

- Completed training on “Machine Learning with Python” under the Enhancing Digital Government and Economy (EDGE) Project of Bangladesh Computer Council, ICT Division.

Walton Hi-Tech Industries PLC

Kaliakoir, Gazipur, Bangladesh

Industrial Trainee, Dept. of Production Management (Refrigerator)

April 2024

- During this immersive experience, we delved into various production processes, including PCB manufacturing, hardware design, SMT line operations, and embedded software engineering.

AWARDS AND SCHOLARSHIPS

- **Dean’s Award**, Khulna University of Engineering and Technology (KUET) – For scoring above 3.75 cgpa in the fourth year.
- **Technical Scholarship**, Khulna University of Engineering and Technology (KUET) – For good academic results from first to third year of undergraduate.
- **General Scholarship**, Dhaka Board – For outstanding result in board exams.

EXTRACURRICULAR ACTIVITIES

Served as General Secretary and Vice-president in different university clubs and organized numerous inter-university and national events.